What Makes a Good Theory?

Weekly Readings

```
schedule <- readODS::read_ods("schedule2.ods")
cat(paste0("* ", strsplit(schedule$Reading[3], split = "\n")[[1]]), sep = "\n")
```

- Frankenhuis, W. E., Panchanathan, K., & Smaldino, P. E. (2023). Strategic Ambiguity in the Social Sciences. Social Psychological Bulletin, 18, 1–25. https://doi.org/10.32872/spb.9923
- Healy, K. (2017). Fuck nuance. Sociological Theory, 35(2), 118-127.

Tutorial

Ambiguity Battle (30 min, 2x 15)

Collaborate with 4 students, divided into two duos: Theorists VS Challengers. One duo chooses one of the following phenomena and writes down a one-sentence theory to explain the phenomenon:

- Venus effect: most people may hold beliefs that are inconsistent with observable phenomena
- Social facilitation: people do better on tasks when they are with other people rather than when they are doing the task alone.
- The frequency illusion: people notice a specific concept, word, or product more frequently after recently becoming aware of it.
- Barnum effect: individuals give high accuracy ratings to descriptions of their personality that supposedly are tailored specifically to them, yet which are in fact vague and general enough to apply to a broad range of people. This effect can provide a partial explanation for the widespread acceptance of astrology, fortune telling, aura reading, and the Myers-Briggs "personality test".
- The bystander effect: individuals are less likely to offer help to a victim in the presence of other people.
- Dunning–Kruger effect: people with low ability in a specific area systematically give overly positive assessments of their ability, while high performers to underestimate their skills.
- The Group Project Paradox: Everyone in the group feels that they did disproportionally more work than the others.

The other duo brainstorms on "challenges" to the theory, for example:

- Can it explain a null result?
- Can your theory explain cross-cultural variation?
- Can it explain developmental change?
- Can it explain contradictory results in experimental versus observational/naturalistic settings?

You may also describe a specific challenge, for example "My study found that people with high agreeableness do not display the Dunning-Kruger effect".

After each challenge, the first group may revise their theory - but without violating the constraint that it must be a single sentence.

The theorists win if the challengers cannot come up with new challenges. The challengers win if the theorists cannot ammend their theory while respecting the one-sentence constraint.

Switch sides after 10 minutes.

Abstraction VS Nuance (30 min)

Starting with the one-sentence theories from the previous exercise, go around in a circle.

1. In turn, each student mentions their personal experience in relation to the theory (so: 3 or 4 rounds per theory).
2. The other students vote on whether this is consistent with the theory, or whether the theory should be made "more nuanced" to accommodate it.
3. All together, amend the theory if necessary.

Do this for 10 minutes, so for 2-3 theories. Then discuss:

- When did each theory stop becoming better?
- Did any additions genuinely improved explanation?
- Which merely increased complexity?

Ambiguous Causal Claims (30 min)

Causal claims are an important part of (social scientific) theories. However, there are strong norms against explicit discussion of causality in the social science literature. The result is that causal language is often ambiguous. @norouziCapturingCausalClaims2025 showed that it is possible to automatically extract causal sentences from published papers despite this ambiguity.

The sentences below are examples from te research by Norouzi and colleagues.

With your group, decide for each sentence and keep notes on:

a. Whether it is Causal (C) or Non-causal (NC)
b. Whether it is Ambiguous (Am) or Straightforward (S)
c. Whether it is Nuanced (N) or Abstract (Ab)

- We find differences in the behavior of our experimental participants across the three countries, and we find that Hofstede's dimensions help to explain these differences.
- In the standard symmetric independent private values model in which $\omega_i(d_i) = 0$, Myerson's (1981) revenue equivalence result shows that, given that the charity never keeps the prize, it maximizes its revenue if it always allocates the prize to the donor with the highest marginal revenue.
- First, models of information signaling have shown that sequential giving can have beneficial effects for voluntary public good provision when the common value of the public good is uncertain.
- Even though the level of regulation a facility faces does not depend on the overall size of a facility, the two factors are highly correlated.
- For example, in social dilemmas with random noise, people lack the ability to judge with certainty if others intended to cooperate or defect (Klapwijk & Van Lange, 2009;Van Lange, Ouwerkerk, & Tazelaar, 2002).
- The presence of a threshold constraining participants' activities will lead participants to be less holistic in their cognitive orientations than when a threshold is not present.
- This result is consistent with a previous study which suggests that behaving dishonestly leads to the forgetting of rules.
- Violent media may cause viewers to perceive greater anger and less gratitude during a cooperative task and as such be less inclined to cooperate.

Solution

- Violent media may cause viewers to perceive greater anger and less gratitude during a cooperative task and as such be less inclined to cooperate. C, S, Ab
- This result is consistent with a previous study which suggests that behaving dishonestly leads to the forgetting of rules. C, S, Ab
- Even though the level of regulation a facility faces does not depend on the overall size of a facility, the two factors are highly correlated. NC, S, Ab
- We find differences in the behavior of our experimental participants across the three countries, and we find that Hofstede's dimensions help to explain these differences. NC, S, N
- The presence of a threshold constraining participants' activities will lead participants to be less holistic in their cognitive orientations than when a threshold is not present. C, Am, N
- First, models of information signaling have shown that sequential giving can have beneficial effects for voluntary public good provision when the common value of the public good is uncertain. C, Am, N
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Discuss:

- How would you go about writing about causality in your work, after having seen these examples?
- What do you think is the difference between ambiguity and nuance?

Homework

Create a graph, annotated with information about functional form

This week, you will extract IF-THEN statements from the sources you selected last week.

Credit: *This assignment is based on the Proposition Based Theory Specification instructions [@glocknerPBTSGuidelinesTemplates2025].*

PBTS Step 2

Individually, fill out Table 2. Translate each source sentence into a set of implications (i.e., IF-THEN statements) and all included propositions that together fully describe the theory.

a. Propositions are statements concerning measurable and non-measurable concepts that can be true or false, and represent an implication's antecedence and consequence. E.g., "A person's needs are satisfied" is a proposition.
b. Propositions can be combined by any logical or mathematical operators, qualifiers, or, if necessary, user-defined uncommon operators (that have to be defined and operationalized).
c. It is often possible to make certain conditions of a theory either part of the propositions or the definition/operationalizations. To keep the propositions simple, these conditions should be part of the definition/operationalizations if they do not belong to the core causal relations of the theory.
d. In this step, remain as close as possible to the original verbal narrative while interpreting the text in accordance with the authors' intended meaning.

(f) In the "Reference" column, reference the label of the text snippet from last week's Table, e.g. S1, S2 … Sn.

g. If the original specification mentions the functional form of a relationship, put this information in the Comment column.

Combining Specifications-BINGO

As a group, read each other's specifications. Select one student's work to be the "starting point" for the definitive version. The author of this version reads each of their implications aloud. Other students put a check mark in front of their implications if these match the implication read by the first student. If they are clearly similar but there are important differences - discuss these and find agreement on which version of the implication to keep in the definitive version.

When the first student is done reading all of their implications, the other students may have some remaining un-checked implications. That is to say: Implications that they coded, but the first student did not. Discuss these to check whether they should be included in the definitive version or not.

The end product should be a complete table of all unique implications that all students (more or less) agree upon.

If there are important disagreements - keep notes, because this is good material for the Discussion section of your final report.

Roll Your Own Theory

If you are making your own theory, do the assignment as described above - but instead of coding selected sentences, each student lists all implications that *they think should be part of the theory*. The group activity then constitutes of discussing these and finding agreement about a set of implications.